DEPARTMENT OF CIVIL ENGINEERING

ENGINEERING SKILL PRACTICE LABORATORY

(CIVIL)

I YEAR- II SEMESTER

REGULATION 2025

NAME: _____

REGISTER NUMBER: _____

YEAR/ SECTION: _____

LABORATORY INSTRUCTION

DO'S AND DON'TS

1. Students shall attend the laboratory sessions punctually and regularly.
2. A minimum of 75% attendance is mandatory to appear for the practical examination.
3. Students shall wear proper lab uniform and closed footwear during lab hours.
4. Use of personal protective equipment such as Shoes, goggles, gloves, and masks is compulsory where required.
5. Students shall strictly follow the instructions given by the lab instructor and technician.
6. Silence and discipline shall be maintained inside the laboratory at all times.
7. Students must read and understand the experiment procedure before starting the work.
8. Experiments shall be carried out only in the allotted groups.
9. Tools and equipment shall be handled carefully and used only for their intended purpose.
10. Students shall not operate any machine or equipment without prior permission.
11. Observations shall be recorded neatly and legibly in the lab record.
12. Lab records must be updated regularly and signed by the instructor.
13. Any accident, injury, or equipment damage shall be reported immediately.
14. All equipment and power supply shall be switched off after completing the experiment.
15. The work area shall be cleaned and tools returned to their designated places before leaving.
16. Students shall bring their lab record and required instruments for every session.
17. Use of mobile phones, eating, or drinking inside the laboratory is strictly prohibited.
18. Indiscipline, negligence, or malpractice will lead to strict disciplinary action.

LIST OF EXPERIMENTS

S.no	Date	Name of experiment	Mark	Faculty Signature
1		Study of building materials - bricks, cement, steel & aggregates & timber		
2		Introduction to Safety Gear, Site Protocols, and Construction Safety Measures		
3		Construction of brick masonry using English and Flemish bond		
4		Measurement of Area of Given Land of Irregular Shape Using Chain Surveying		
5		Compass surveying for bearing measurement and area calculation		
6		Demonstration of Theodolite and Levelling Instruments		
7		Drafting of a simple Building plan using Autocad		
8		Integration of plumbing and sanitation layouts in building drawing		
9		Plumbing Tools and Practices: Pipe Cutting, Threading, Jointing, Installation, and Testing		
10		Mixing and casting of concrete		

Study and Demonstration

of

Civil Engineering Tools, Building Materials, and Safety Practices

Exp.No- 01

STUDY OF BUILDING MATERIALS

BRICKS, CEMENT, STEEL & AGGREGATES & TIMBER

Date:

OBJECTIVES

- To study the basic theory and properties of building materials
- To identify different types of materials used in construction
- To understand standard laboratory tests through demonstration
- To correlate material properties with their practical applications

1: BRICKS

Bricks are artificial stone units made from clay by molding, drying, and burning in kilns. They are widely used in masonry construction due to their durability, strength, and fire resistance. Good quality bricks should be uniform in shape, size, and colour, free from cracks, and should produce a clear ringing sound when struck together. Bricks possess adequate compressive strength and low water absorption to ensure durability.

Properties

- Shape, size, and colour
- Texture and hardness
- Water absorption
- Compressive strength

Uses of Bricks

- Load-bearing and partition walls
- Piers and columns
- Pavements and drains

2: CEMENT

Cement is a binding material which sets and hardens when mixed with water. Ordinary Portland Cement (OPC) is manufactured by grinding clinker formed by burning a mixture of limestone and clay at high temperature. Cement is responsible for strength development in concrete. Its properties such as fineness, setting time, soundness, and strength affect the quality of construction. Grade of cement – 33, 43 & 53. While ancient civilizations used early forms of cement, modern Portland cement, the basis for today's concrete, was invented by English bricklayer Joseph Aspdin in 1824.

Properties

- Colour and fineness
- Standard consistency
- Initial and final setting time
- Soundness

Types of Cement

- Ordinary Portland Cement (OPC)
- Portland Pozzolana Cement (PPC)
- Portland Slag Cement (PSC)

3: STEEL

Steel is an alloy of iron and carbon with high tensile strength and ductility. In reinforced cement concrete (RCC) structures, steel is used to resist tensile stresses, while concrete resists compressive stresses. Thermo-Mechanically Treated (TMT) bars are commonly used in construction due to their high strength, ductility, and corrosion resistance.

Properties

- Diameter and surface pattern
- Yield strength
- Tensile strength
- Ductility

Uses of Steel

- Reinforcement in RCC structures
- Structural members such as beams and columns

4: AGGREGATES

Aggregates are inert granular materials such as sand, gravel, and crushed stone. They occupy about 70–80% of the volume of concrete and provide bulk, strength, and durability. Aggregates are classified as fine aggregates (sand) and coarse aggregates (gravel or crushed stone). Proper grading of aggregates is essential for producing strong and durable concrete

Properties

- Size and shape
- Surface texture
- Grading of aggregates
- Cleanliness

Uses of Aggregates

- Concrete and mortar
- Road construction
- Railway ballast

5: TIMBER

Timber is a natural building material obtained from trees. It is commonly used due to its strength, durability, workability, and aesthetic appearance. Good timber should have straight grains, uniform colour, sufficient strength, and resistance to decay and insects. Timber is seasoned before use to reduce moisture content and improve durability.

Properties Studied

- Colour and grain structure
- Moisture content
- Strength and durability
- Defects in timber

Common Types of Timber

- Teak
- Sal
- Deodar
- Pine

Uses of Timber

- Doors and windows
- Formwork
- Furniture and flooring

CONCLUSION

This study/demonstration manual provides theoretical knowledge and basic understanding of building materials, helping students connect classroom learning with practical applications in civil engineering.

Experiment No.: 2

Introduction to Safety Gear,

Site Protocols, and Construction Safety Measures

Date:

Aim

To familiarize students with personal protective equipment (PPE), construction site safety protocols, and basic safety measures followed at civil engineering construction sites.

Objectives

- To understand the importance of safety in construction sites
- To identify various types of safety gear used in civil engineering works
- To learn standard site safety protocols and hazard prevention methods
- To create awareness about accident prevention and safe work practices

Apparatus / Materials Required

- Safety helmet
- Safety shoes
- Reflective safety jacket
- Safety goggles
- Hand gloves
- Face mask / respirator
- Ear protection (ear plugs / ear muffs)
- Safety signage charts

Theory

Construction sites involve activities such as excavation, concreting, lifting operations, and working at heights, which pose serious safety risks. Proper use of **personal protective equipment (PPE)** and adherence to **site safety protocols** are essential to prevent accidents, injuries, and fatalities.

Safety measures in construction include:

- Use of appropriate safety gear
- Following site entry and exit protocols
- Awareness of hazards and warning signs
- Safe handling of tools, machinery, and materials
- Emergency preparedness and first-aid practices

Safety management ensures compliance with statutory regulations and promotes a safe working environment for engineers, supervisors, and workers.

Safety Gear and Their Uses

Safety Gear	Purpose
Safety Helmet	Protects head from falling objects
Safety Shoes	Prevents foot injuries from sharp/heavy objects
Safety Goggles	Protects eyes from dust and debris
Hand Gloves	Prevents cuts, burns, and chemical exposure
Safety Jacket	Improves visibility at site
Face Mask	Protects from dust and harmful particles
Ear Protection	Reduces noise-related hearing damage

Construction Site Safety Protocols

- Mandatory use of PPE inside the site
- Restricted access to unauthorized personnel
- Proper display of safety and warning signs
- Safe storage of materials and tools
- Regular inspection of equipment and machinery
- Emergency exits and first-aid availability

Precautions

- Safety gear shall be worn properly and consistently.
- Damaged or defective PPE shall not be used.
- Unsafe acts and shortcuts shall be strictly avoided.
- Safety instructions given by supervisors shall be followed.

Result

Thus, the students were familiarized with various safety gear**,** site protocols, and construction safety measures followed in civil engineering construction practices.

Masonry and Finishing Works

Experiment No.: 3

Construction of brick masonry using

English and Flemish bond

Date:

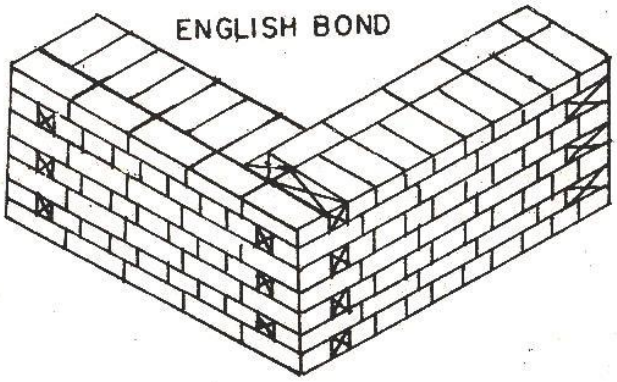
Aim: To construct brick masonry using English bond and Flemish bond and to understand their arrangement, strength characteristics, and applications in construction.

Objectives

- To identify different brick bonds used in masonry construction
- To learn the procedure for constructing English and Flemish bond masonry
- To understand the importance of bonding in strength and load distribution
- To gain hands-on experience in brick laying and mortar application

Materials / Tools Required

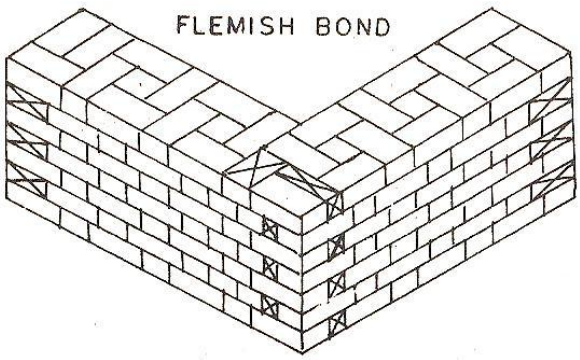
- Burnt clay bricks
- Cement
- Sand
- Water
- Brick trowel
- Spirit level
- Plumb bob
- Line and pins
- Measuring tape
- Mortar pan



$1\frac{1}{2}$ BRICK WALL

Theory

Brick masonry is constructed by laying bricks bonded together with mortar. **Bonding** ensures proper load transfer, strength, and stability of masonry structures.



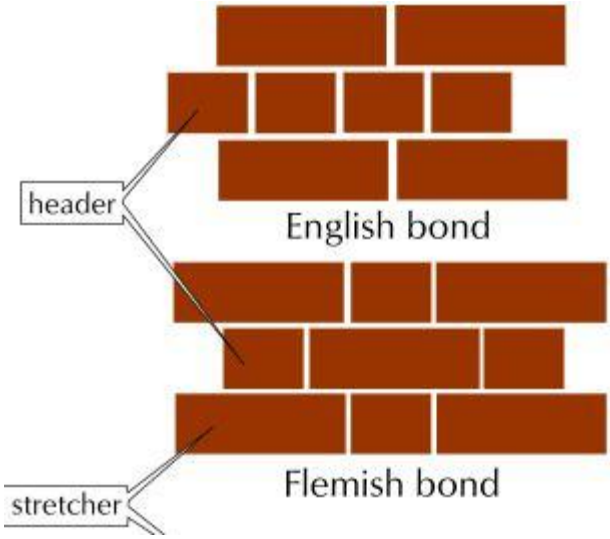
$1\frac{1}{2}$ BRICK WALL

English Bond

- Consists of **alternate courses of headers and stretchers**
- Strongest and most commonly used bond
- Suitable for load-bearing walls

Flemish Bond

- Each course consists of **alternate headers and stretchers**
- More aesthetic appearance than English bond
- Comparatively weaker; often used in residential and architectural works



Procedure

A. Construction of English Bond

1. Prepare cement mortar of required proportion.
2. Lay a mortar bed on the base.
3. Place the first course using header bricks with proper alignment.
4. Lay the next course using stretcher bricks.
5. Maintain uniform mortar thickness (about 10 mm).
6. Check verticality using plumb bob and horizontality using spirit level.

B. Construction of Flemish Bond

1. Prepare the mortar and base as before.
2. Lay the first course with alternate header and stretcher bricks.
3. Ensure headers are centred over stretchers in the course below.
4. Maintain proper alignment and joint thickness.
5. Repeat the same pattern for subsequent courses.

Observations

1. Alignment of bricks
2. Uniformity of mortar joints
3. Proper bonding pattern
4. Verticality and level of masonry

Precautions

- Bricks shall be soaked in water before use.
- Mortar consistency shall be uniform.
- Joints shall not exceed recommended thickness.
- Proper bonding pattern shall be strictly followed.

Result

Thus, brick masonry was successfully constructed using English bond and Flemish bond, and their bonding patterns and characteristics were studied.

Basic Surveying Practices

Experiment No.: 4

Measurement of Area of Given Land of Irregular Shape Using Chain Surveying (Heron's Formula)

Date:

Aim:
To measure the area of an irregular plot of land using chain surveying by dividing it into triangles and applying Heron's Formula.

Apparatus Required:
1. Chain (20m or 30m)
2. Arrows (10)
3. Ranging rods
4. Pegs or wooden stakes
5. Plumb bob and cross-staff (if needed)
6. Field book, notebook, and pencil
7. Calculator

Procedure:
1. **Reconnaissance Survey:**
Walk around the plot and observe the boundaries.
Identify and mark the corners of the plot with pegs or ranging rods.

2. **Station Setup and Triangulation:**
Divide the irregular plot into a number of triangles by joining suitable points.
Name all stations (A, B, C, D…).

3. **Chain Measurement:**
Measure the lengths of all sides of the triangles using the chain.
Use ranging rods to ensure straight alignment while chaining.
Record all side lengths accurately in a field book.

4. **Area Calculation Using Heron's Formula:**
For each triangle:
Let the triangle have sides a, b, and c.
Compute the semi-perimeter (s):
$$s = (a + b + c) / 2$$
Calculate the area (A):
$$A = \sqrt{[s(s - a)(s - b)(s - c)]}$$
Repeat for all triangles and sum them to get the total area.

Observation Table:

Triangle No.	Side a (m)	Side b (m)	Side c (m)	Semi-perimeter s (m)	Area (sq. m)
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Result:
The total area of the given irregular plot using chain surveying and Heron's formula = _____ sq. meters.

Precautions:

- Ensure the chain is straight and horizontal while measuring.
- Avoid sagging and record readings accurately.
- Select triangles that are well-conditioned.
- Repeat doubtful measurements for accuracy.

Experiment No.:5

Compass surveying for bearing measurement and area calculation

Date:

Aim:

To compute the interior angle at a selected station, point on a land area using a prismatic compass by measuring the magnetic bearings of the lines meeting at that point.

2. Apparatus Required:

S. No	Apparatus	Quantity
1	Prismatic Compass	1
2	Tripod Stand	1
3	Ranging Rods	3
4	Survey Field Book	1
5	Pegs and Hammer	As needed
6	Measuring Tape (optional)	1
7	Plumb Bob	1
8	Pencil and Eraser	-

3. Theory:

A prismatic compass is a surveying instrument used to measure the magnetic bearing of lines. Bearings are measured in whole-circle bearing (WCB) system from 0° to 360° in a clockwise direction from the magnetic north.

The interior angle at a point is the angle formed between two survey lines that converge at that point. It can be calculated by taking the difference between the magnetic bearings of the two lines.

4. Procedure:

1. Visit the site and select three or more stations forming a polygon (e.g., triangle ABC). Mark the stations A, B, and C.
2. Set up the prismatic compass on a tripod over the selected station (e.g., B) using a plumb bob and level it properly.
3. Sight towards station A and record the bearing of line AB.
4. Sight towards station C and record the bearing of line BC.
5. Repeat if necessary at other points. Use the bearings to compute the interior angle at B.

5. Formula Used:

Interior Angle (θ) at B = (Bearing of BC) – (Bearing of BA)
If result is negative, add 360°.

θ = | Bearing2 – Bearing1 |
If θ > 180°, Interior Angle = 360° – θ

6. Sample Calculation:

Assume:
Bearing of AB = 45°
Reverse bearing of AB (i.e., BA) = 45° + 180° = 225°
Bearing of BC = 160°
θ = |160° - 225°| = 65° (Since negative, add 360°, then take 360° - 295° = 65°)
So, Interior Angle at B = 65°

7. Tabulation Format:

Station	Line Observed	Fore Bearing (°)	Back Bearing (°)	Interior Angle (°)
	-	-	-	

8. Result:

The interior angle at different station of given land was calculated using prismatic compass readings.

Experiment No.: 6

Demonstration of Theodolite and

Levelling Instruments

Date:

Aim

To demonstrate the construction, parts, working principles, and uses of theodolite and levelling instruments used in surveying.

Objectives

- To identify the main parts of a theodolite and leveling instrument
- To understand their basic working principles
- To learn the procedures for setting up and leveling the instruments
- To understand applications in civil engineering surveying

Instruments Required

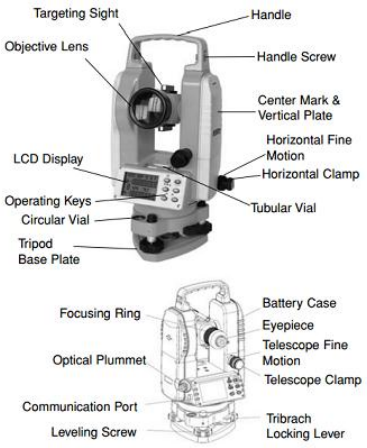
- Theodolite
- Dumpy level / Auto level
- Tripod stand
- Leveling staff
- Plumb bob

Theory

- Surveying instruments are used to determine **relative positions, elevations, and angles** on the earth's surface.

Theodolite

A theodolite is a precision instrument used for **measuring horizontal and vertical angles**. It is widely used in triangulation, alignment of structures, and route surveying.



Levelling Instrument

A leveling instrument is used to determine **difference in elevation** between points. Dumpy level and auto level are commonly used in construction works such as roads, buildings, and canals.

Parts and Functions

Main Parts of Theodolite

- Telescope – used for sighting the object
- Horizontal circle – measures horizontal angles
- Vertical circle – measures vertical angles
- Levelling screws – used for levelling the instrument
- Tripod – supports the instrument

Main Parts of Levelling Instrument

- Telescope
- Spirit level / compensator
- Foot screws
- Tripod stand
- Levelling staff (used for staff readings)

Procedure

A. Demonstration of Theodolite

1. Fix the theodolite on the tripod.
2. Center the instrument using plumb bob.
3. Level the instrument using foot screws.
4. Demonstrate horizontal and vertical angle measurement.
5. Explain temporary adjustments and sighting method.

B. Demonstration of Levelling Instrument

1. Fix the level on the tripod at a suitable location.
2. Level the instrument using foot screws.
3. Focus the eyepiece and objective.
4. Take staff readings on different points.
5. Explain concept of back sight, fore sight, and reduced levels.

Observations

- Proper leveling of instruments
- Clarity of staff readings
- Smooth movement of telescope
- Accurate alignment and focusing

Result

Thus, the construction, parts, working principles, and applications of theodolite and levelling instruments were demonstrated and understood.

Building Drawing

and

Plumbing Layout Preparation

Experiment No.: 7

Drafting of a simple Building plan using Autocad

Date:

Aim

To prepare a **simple residential building plan** using **manual drafting / AutoCAD**, showing proper room layout, dimensions, doors, windows, and orientation as per standard building drawing practices.

Apparatus / Software Required

For Manual Drafting:

- Drawing board
- Drawing sheet (A3 or A2 size)
- Pencils (HB, 2H)
- Eraser and scale

For AutoCAD Drafting:

- Computer system
- AutoCAD software (2018 or above)
- Mouse and keyboard

Building Details (Given Data)

- **Type of building:** Single-storey residential building
- **Plot size:** 10 m × 8 m
- **Built-up area:** 70 m² (approx.)
- **Accommodation:**
  - 1 Living room
  - 1 Kitchen
  - 2 Bedrooms
  - 1 Toilet
  - 1 Bathroom
  - 1 Balcony / Sit-out

Standard Room Dimensions

Room	Size (m × m)
Living Room	4.0 × 3.5
Bedroom 1	3.5 × 3.0
Bedroom 2	3.0 × 3.0
Kitchen	3.0 × 2.5
Toilet	1.5 × 1.2
Bathroom	1.5 × 1.2
Balcony	1.2 wide

Procedure

A. Manual Drafting Procedure

1. Fix the drawing sheet on the drawing board.
2. Draw a **border line** and **title block**.
3. Select a suitable scale (e.g., **1:100**).
4. Draw the **outer walls** with a thickness of **230 mm**.
5. Draw **internal partition walls** with **115 mm** thickness.
6. Mark the room layout as per dimensions.
7. Draw **doors and windows** using standard symbols.
8. Provide **room names and dimensions**.
9. Add **north direction symbol** and scale note.

AutoCAD Drafting Procedure

1. Open AutoCAD → Set **units** to meters.
2. Create layers:
   - Walls
   - Doors
   - Windows
   - Text
   - Dimensions

3. Draw outer and inner walls using **LINE / OFFSET** commands.
4. Insert doors and windows using **BLOCKS**.
5. Apply **DIMLINEAR** for dimensions.
6. Label rooms using **TEXT / MTEXT**.
7. Add **north symbol** and **scale**.
8. Save the drawing and plot at scale **1:100**.

Drawing Description

The plan consists of a centrally located living room with two bedrooms on either side. The kitchen is placed adjacent to the living room with easy access. Toilet and bathroom are provided separately for hygienic purposes. Proper ventilation is ensured through windows, and a balcony is provided at the front for natural lighting and airflow.

Result

A **simple residential building plan** was successfully drafted using **manual drafting / AutoCAD**, clearly showing room layout, dimensions, doors, windows, and orientation as per standard drafting practice.

Experiment No.: 8

Integration of plumbing and sanitation layouts in building drawing

Date:

AIM

To prepare a **building drawing integrated with plumbing and sanitation layouts**, showing water supply lines, sanitary fixtures, drainage pipes, inspection chambers, and connection to municipal sewer/septic tank as per standard civil engineering practice.

OBJECTIVES

- To understand the **coordination between architectural planning and plumbing systems**.
- To learn the **layout of water supply and sanitary drainage** in a residential building.
- To prepare plumbing and sanitation drawings using **standard symbols and dimensions**.
- To ensure **hygienic, functional, and efficient water management** in buildings.

APPARATUS / SOFTWARE REQUIRED

Manual Drafting:

- Drawing board, drawing sheet (A3/A2)
- Scale
- Pencils (HB, 2H), eraser

AutoCAD Drafting:

- Computer system
- AutoCAD software
- Mouse and keyboard

4. BUILDING DETAILS

- **Type of building:** Residential (single storey)
- **Accommodation:**
  - Living room
  - Kitchen
  - 2 Bedrooms
  - Toilet & Bathroom
- **Water source:** Municipal water supply
- **Sanitation system:** Underground drainage with inspection chambers

STANDARD PLUMBING AND SANITATION COMPONENTS USED

Sanitary Fixtures:

- Water closet (WC)
- Wash basin
- Shower
- Kitchen sink

Plumbing Elements:

- Overhead water tank
- Down-take pipes
- Water supply pipes (15 mm, 20 mm)
- Soil pipe (100 mm)
- Waste pipe (75 mm)
- Vent pipe (50 mm)
- Floor trap
- Gully trap
- Inspection chamber

STANDARD PIPE SIZES

Component	Diameter
Water supply pipe	15–20 mm
Soil pipe (WC)	100 mm
Waste pipe	75 mm
Vent pipe	50 mm
Floor trap outlet	75 mm

PROCEDURE

A. Building Plan Preparation

1. Draw the **architectural plan** showing rooms, walls, doors, and windows.
2. Maintain proper **scale (1:100)** and wall thickness.
3. Mark toilet, bathroom, and kitchen locations clearly.

B. Plumbing Layout

1. Mark the **overhead tank** position.
2. Draw **water supply lines** to kitchen, bathroom, WC, and wash basin.
3. Show **valves and pipe sizes** using standard symbols.
4. Ensure minimum pipe crossings and easy accessibility.

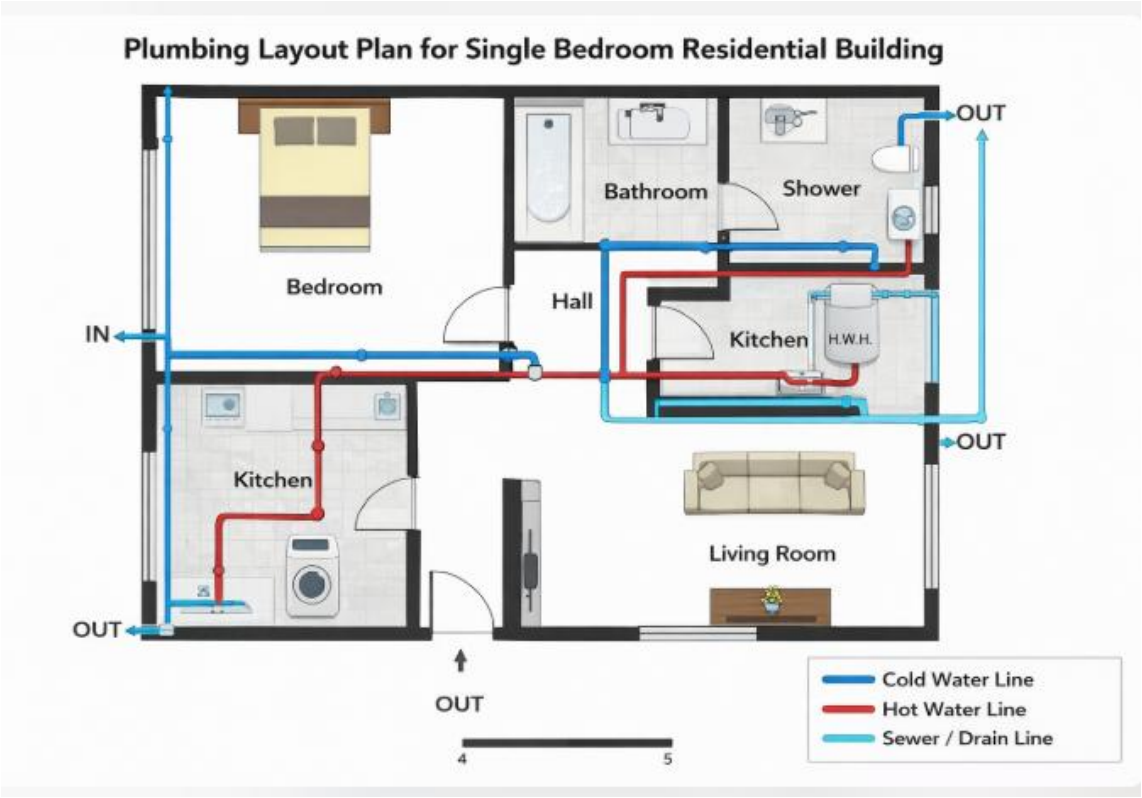
C. Sanitation Layout

1. Draw **soil and waste pipes** from toilets and bathrooms.
2. Provide **floor traps** in bathrooms and kitchens.
3. Connect waste pipes to **gully traps**.
4. Show **inspection chambers** at junctions and direction changes.
5. Connect the system to **municipal sewer / septic tank**.
6. Provide **vent pipes** for proper gas release

DRAWING DETAILS

- Plumbing layout plan
- Sanitation layout plan
- Standard plumbing symbols

- Pipe diameters and flow direction arrows
- North direction and scale



RESULT

The **plumbing and sanitation layouts were successfully integrated** with the building drawing, ensuring proper water supply, effective wastewater disposal, and compliance with standard civil engineering practices.

Plumbing Tools and Practices

Experiment No.: 9

Plumbing Tools and Practices: Pipe Cutting, Threading, Jointing, Installation, and Testing

Date:

Aim

To identify common plumbing tools and materials, demonstrate pipe cutting, threading, and jointing for PVC and GI pipes, install a basic plumbing system, and conduct leakage and pressure testing.

Objectives

1. To identify plumbing tools and materials used in building services
2. To learn pipe cutting, threading, and jointing techniques
3. To install a basic plumbing layout (tap, basin, drainage)
4. To test the system for leakage and pressure

Tools and Materials Required

Tools

- Pipe cutter / hacksaw
- Pipe vice
- Die and stock (for GI threading)
- Pipe wrench / spanner
- Measuring tape
- Plumber's reamer
- Screwdriver

Materials

- PVC pipes and fittings (elbow, tee, socket)
- GI pipes and fittings
- PVC solvent cement
- Thread sealing tape / hemp
- Wash basin
- Tap and valves
- Flexible hose / trap
- Water supply source

Theory

Plumbing systems are essential for **water supply and sanitation** in buildings. Proper selection of materials, accurate pipe cutting, correct jointing, and systematic installation ensure **leak-free and durable systems**.

- **PVC pipes** are lightweight, corrosion-resistant, and commonly used for water supply and drainage.
- **GI pipes** are strong and used where higher-pressure resistance is required. Testing ensures the system can withstand operating pressure without leakage.

Procedure

A. Identification of Plumbing Tools and Materials

1. The instructor displays common plumbing tools and fittings.
2. Students identify each tool and note its function.
3. Differences between PVC and GI pipes are explained.

B. Pipe Cutting

1. Measure the required pipe length using measuring tape
2. Cut PVC pipe using pipe cutter/hacksaw.
3. Remove burrs using reamer for smooth joints.

C. Threading and Jointing

GI Pipe Threading

1. Fix the GI pipe in the pipe vice.
2. Apply cutting oil and use die and stock to cut threads.
3. Clean threads and apply sealing tape.
4. Join fittings tightly using pipe wrench.

PVC Pipe Jointing

1. Clean pipe ends and fittings.
2. Apply solvent cement uniformly.
3. Insert pipe into fitting and hold for proper setting.

D. Installation of Basic Plumbing System

1. Fix the wash basin at proper height.
2. Connect water supply line to the tap.
3. Connect basin outlet to drainage pipe with trap.
4. Ensure proper alignment and support for pipes.

E. Testing for Leakage and Pressure

1. Supply water to the installed system.
2. Check all joints visually for leakage.
3. Apply operating pressure and observe performance.
4. Rectify leaks if found and retest.

Result

Thus, plumbing tools and materials were identified, **pipe cutting, threading, and jointing** were demonstrated, a **basic plumbing system** was installed, and **leakage and pressure testing** were successfully performed.

Concrete Work and Field Exposure

Experiment No.: 10
Mixing and casting of concrete

Date:

1. Aim

To prepare concrete by proper proportioning, mixing and casting.

2. Apparatus Required

- Cement (Ordinary Portland Cement – OPC)
- Fine aggregate (sand)
- Coarse aggregate (gravel)
- Clean potable water
- Weighing balance
- Measuring jar / bucket
- Mixing tray (non-absorbent)
- Trowel and shovel
- Concrete moulds (cube: 150 × 150 × 150 mm)
- Vibrating table or tamping rod
- Steel float
- Oil or grease (for moulds)

3. Working Principle

Concrete is a composite material formed by mixing cement, fine aggregate (FA), coarse aggregate (CA) and water in suitable proportions. Cement acts as a binding material. When water is added, hydration starts, forming a paste that coats the aggregates and hardens with time, resulting in a strong and durable material. The strength and workability of concrete depend on 1. Water–cement ratio 2. Type and grading of aggregates 3. Method of mixing and compaction

4. Types of Concrete Mix

4.1 Nominal Mix

It is used for small-scale works where strength requirement is not critical.

Grade	Mix Proportion (C : FA : CA)
M5	1 : 5 : 10
M10	1 : 3 : 6
M15	1 : 2 : 4
M20	1 : 1.5 : 3

5.2 Design Mix

Proportions are calculated based on strength, durability and workability requirements using IS 10262.

5.3 Ready-Mix Concrete (RMC)

Concrete prepared in batching plants and transported to site.

6. Procedure

6.1 Preparation

1. Clean the mixing tray and tools.
2. Apply oil to the inner surfaces of the moulds.
3. Measure materials accurately using a weighing balance.

6.2 Mixing of Concrete (Manual Mixing)

1. Spread measured fine aggregate evenly on the mixing tray.
2. Add cement over the sand and mix dry until uniform color is obtained.
3. Add coarse aggregate and mix thoroughly.
4. Make a crater at the center and add required quantity of water gradually.
5. Mix continuously until a homogeneous and workable mix is achieved.

6.3 Casting of Concrete

1. Fill the concrete mould in three equal layers.
2. Compact each layer using a tamping rod (25 strokes per layer).
3. Level the top surface using a trowel.
4. Mark the specimen for identification.

6.4 Curing

- Leave the mould undisturbed for 24 hours.
- Remove the specimen and cure it in water for the required period (7, 14, or 28 days).

8. Model Calculations

(Grade of concrete = M20 Mix proportion = 1 : 1.5 : 3)

Step 1: Total proportion

Total = 1 + 1.5 + 3 = 5.5

Step 2: Quantity of materials for 1 m³ of concrete

Volume of dry concrete = 1.54 m³

Cement:

[Cement = (1 / 5.5) × 1.54 = 0.28 m^3]

Weight of cement:

[0.28 × 1440 = 403 kg ≈ 8 bags]

Fine Aggregate:

[(1.5 / 5.5) × 1.54 = 0.42 m^3]

Coarse Aggregate:

[(3 / 5.5) × 1.54 = 0.84 m^3]

Water:

[Water = 0.5 × weight of cement = 0.5 × 403 = 201.5 liters]

9. Formulae Used

- Volume of dry concrete = 1.54 × wet volume
- Water–cement ratio = Weight of water / Weight of cement
- Cement weight = Volume × Density of cement
- Density of cement = 1440 kg/m^3

10. Observations

- Concrete mix was uniform and cohesive
- No segregation observed
- Proper compaction improved surface finish

11. Result

Concrete was successfully mixed and cast using the specified mix proportion, producing a homogeneous and workable concrete specimen suitable for strength testing.